Lab 9: Wireless Network Configuration and Packet Analysis using Cisco Packet Tracer

Objectives

- To configure a wireless router and establish a working wireless network using Cisco Packet Tracer.
- To connect multiple wireless devices and assign IP addresses automatically through DHCP.
- To implement WPA2 security to protect wireless communication.
- To monitor and examine packet transmission using Simulation Mode in Packet Tracer.

Theory

Fundamentals of Wireless Networking

Wireless networking enables devices to communicate without physical cables by using radio frequency (RF) signals. A wireless router serves as the main connecting device, allowing laptops, smartphones, and other wireless clients to access the network. Wi-Fi communication is governed by IEEE 802.11 standards, ensuring compatibility between devices.

Functions of a Wireless Router

A wireless router performs three primary roles within a network environment:

- Access Point (AP): Connects wireless client devices to the network.
- Router: Transfers data between the local network and external networks.
- DHCP Server: Automatically distributes IP addresses to connected devices.

In Cisco Packet Tracer, the WRT300N router model is typically used to simulate wireless network configurations.

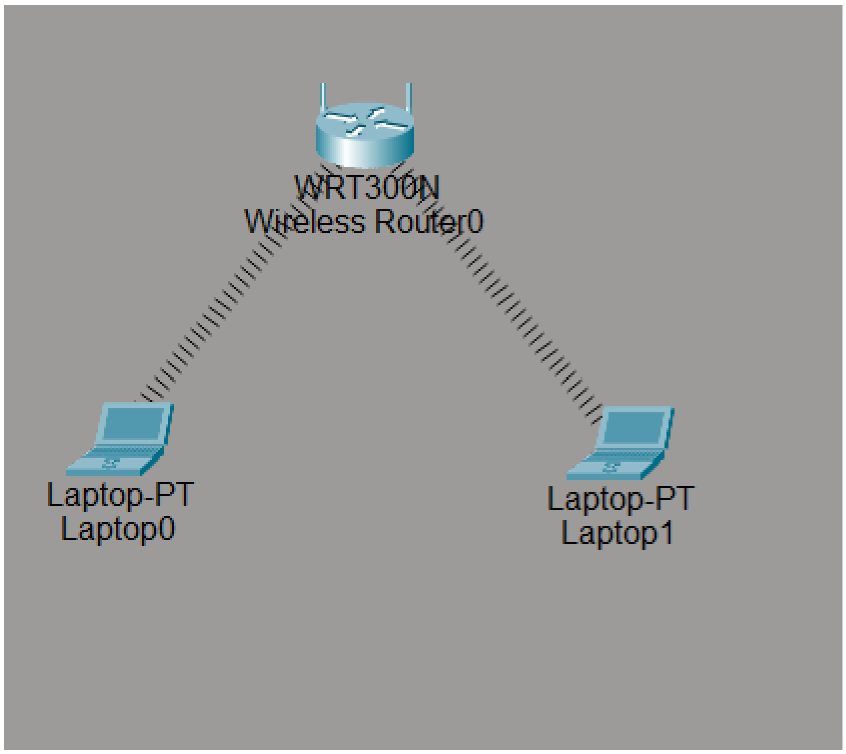
Wireless Security – WPA2

Unsecured wireless networks are vulnerable to unauthorized access. WPA2 (Wi-Fi Protected Access 2) is a widely adopted security protocol that utilizes AES (Advanced Encryption Standard) encryption to secure data transmission. Implementing WPA2 ensures proper user authentication, maintains data confidentiality, and protects against threats such as eavesdropping and unauthorized intrusion.

Packet Analysis in Cisco Packet Tracer

Simulation Mode in Packet Tracer allows users to visualize how data packets travel through the network. Through packet analysis, users can observe DHCP request and acknowledgment processes, ARP operations, and encrypted wireless communication under WPA2. This practical observation strengthens understanding of theoretical networking concepts.

Network Topology



The lab utilizes a basic wireless star topology, where the wireless router acts as the central device and all client devices connect directly to it. Communication between laptops occurs through the router, which centrally manages IP addressing and security. This configuration represents a typical home or small office network setup.

Configuration

The DHCP-assigned IP addresses of the laptops are listed below:

Device	IPv4 Address	Subnet Mask	Default Gateway
Laptop0	192.168.0.101	255.255.255.0	192.168.0.1
Laptop1	192.168.0.102	255.255.255.0	192.168.0.1

Wireless Router Configuration Steps

Step 1: Assign Basic Settings

Begin by opening the Wireless Router in Packet Tracer. Navigate to GUI > Setup and enter the following settings:

- IP Address: 192.168.0.1
- Subnet Mask: 255.255.255.0
- DHCP Server: Enabled

Wireless Configuration

Step 2: Enable Basic Wireless Settings

On the router, go to Wireless > Basic Wireless Settings and switch to the following parameters:

- Network Mode: Mixed
- SSID: HCOE
- Channel: Auto

Click Save Settings.

Wireless Security Configuration

Step 3: Enable Wireless Security

On the router, navigate to Wireless > Wireless Security and turn on the security with these details:

- Security Mode: WPA2-Personal
- Encryption: AES
- Passphrase: hcoe@123

Save the configuration.

Observations

Perform a ping test from Laptop0 to Laptop1 to confirm network connectivity.

```
C:\>ping 192.168.0.100

Pinging 192.168.0.100 with 32 bytes of data:

Reply from 192.168.0.100: bytes=32 time=40ms TTL=128
Reply from 192.168.0.100: bytes=32 time=16ms TTL=128
Reply from 192.168.0.100: bytes=32 time=23ms TTL=128
Reply from 192.168.0.100: bytes=32 time=22ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 40ms, Average = 25ms
```

Discussion

The wireless network was successfully established in Cisco Packet Tracer using the WRT300N router and laptop devices. DHCP simplified configuration by automatically assigning IP addresses, minimizing manual errors. WPA2 security ensured encrypted communication between devices, emphasizing the importance of protecting wireless networks. Packet analysis in Simulation Mode verified correct DHCP exchanges and secure data transfer, confirming the practical implementation of theoretical concepts.

Conclusion

This lab demonstrated how to configure a wireless network in Cisco Packet Tracer. It highlighted the role of DHCP in dynamic IP allocation and the significance of WPA2 in securing wireless communication. Packet analysis confirmed proper network functionality, enhancing both conceptual understanding and practical networking skills.